# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 1 | Each point where the graph meets the $x$-axis is a zero |
| 2 | **(a)** 5 | Sum of zeroes $= 7$, so other $= 7 - 2 = 5$ |
| 3 | **(b)** $x^2 - 7x + 12$ | $x^2 - (\text{sum})x + \text{product}$ |
| 4 | **(d)** 9 | $\alpha + \beta = -b/a$ |
| 5 | **(a)** 14 | $\alpha\beta = c/a$ |
| 6 | **(d)** 2 and 3 | $(x-2)(x-3) = 0$ gives $x = 2$ or $x = 3$ |
| 7 | **(d)** 7 | Sum $= -b/a$ |
| 8 | **(b)** 2 | Each point where the graph meets the $x$-axis is a zero |
| 9 | **(c)** 6 | Sum of zeroes $= 9$, so other $= 9 - 3 = 6$ |
| 10 | **(a)** $x^2 - 3x - 10$ | $x^2 - (\text{sum})x + \text{product}$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Zeroes are $-2$ and 3 (1); a parabola, so degree 2 (1). | 1+1 |
| 12 | Sum $= -\dfrac{b}{a} = 4$ (1); product $= \dfrac{c}{a} = 3$ (1). | 1+1 |
| 13 | $x^2 - (\frac{1}{4})x + (-1)$ (1) $= x^2 - \frac{1}{4}x - 1$ (1). | 1+1 |
| 14 | Zeroes are 1 and 4 (1); a parabola, so degree 2 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Zeroes $-2$, 3 (1); sum $1 = -b/a$ (1); product $-6 = c/a$ (1). | 1+1+1 |
| 16 | $p(2) = 0$: $4 - 10 + 6 = 0$ gives $k = -5$ (1); sum of zeroes $= -k = 5$ (1); other zero $= 5 - (2) = 3$ (1). | 1+1+1 |
| 17 | Sum $= 10$, product $= 21$ (1+1); polynomial $x^2 - (10)x + (21) = x^2 - 10x + 21$ (1). | 1+1+1 |
| | **OR:** Zeroes $\frac{1}{2}$, 4 (1); sum $\frac{9}{2} = -b/a$ (1); product $2 = c/a$ (1). | 1+1+1 |
| 18 | $p(3) = 0$: $9 - 21 + 12 = 0$ gives $k = -7$ (1); sum of zeroes $= -k = 7$ (1); other zero $= 7 - (3) = 4$ (1). | 1+1+1 |
| | **OR:** Sum $= 2$, product $= -24$ (1+1); polynomial $x^2 - (2)x + (-24) = x^2 - 2x - 24$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $\alpha + \beta = 7$, $\alpha\beta = 10$ (1+1); new sum $= 9$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = 18$ (1); polynomial $x^2 - (9)x + (18)$ (1). | 1+1+1+1+1 |
| | **OR:** Sum of zeroes $= -b/a = 8$ (2); third zero $= 8 - (1) - (2) = 5$ (2); check by product $= 10$ (1). | 2+2+1 |
| 20 | $\alpha + \beta = 3$, $\alpha\beta = -4$ (1+1); new sum $= 5$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = 0$ (1); polynomial $x^2 - (5)x + (0)$ (1). | 1+1+1+1+1 |
| | **OR:** Sum of zeroes $= -b/a = 6$ (2); third zero $= 6 - (-1) - (3) = 4$ (2); check by product $= -12$ (1). | 2+2+1 |

**Mistakes to avoid:**

- Writing sum of zeros as $\dfrac{c}{a}$ and product as $-\dfrac{b}{a}$; it is sum $= -\dfrac{b}{a}$ and product $= \dfrac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not $-8$.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2 + (\alpha + \beta)x + \alpha\beta$ instead of $x^2 - (\alpha + \beta)x + \alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 5 | $\alpha+\beta=-b/a$ |
| 2 | **(c)** -24 | $\alpha\beta=c/a$ |
| 3 | **(b)** 4 and 1 | $(x-4)(x-1)=0$ gives $x=4$ or $x=1$ |
| 4 | **(d)** 7 | Sum $=-b/a$ |
| 5 | **(d)** 3 | Each point where the graph meets the $x$-axis is a zero |
| 6 | **(b)** 4 | Sum of zeroes $=3$, so other $=3-{-1}=4$ |
| 7 | **(b)** $x^2-9x+8$ | $x^2-(\text{sum})x+\text{product}$ |
| 8 | **(a)** 9 | $\alpha+\beta=-b/a$ |
| 9 | **(c)** 20 | $\alpha\beta=c/a$ |
| 10 | **(c)** 5 and 6 | $(x-5)(x-6)=0$ gives $x=5$ or $x=6$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Sum $=-\frac{b}{a}=-\frac{5}{3}$ (1); product $=\frac{c}{a}=-\frac{2}{3}$ (1). | 1+1 |
| 12 | $x^2-(2)x+(\frac{1}{3})$ (1) $=x^2-2x+\frac{1}{3}$ (1). | 1+1 |
| 13 | Zeroes are $-3$ and 1 (1); a parabola, so degree 2 (1). | 1+1 |
| 14 | Sum $=-\frac{b}{a}=1$ (1); product $=\frac{c}{a}=-\frac{3}{4}$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Zeroes $-5$, 2 (1); sum $-3=-b/a$ (1); product $-10=c/a$ (1). | 1+1+1 |
| 16 | $p(-2)=0$: $4-2-2=0$ gives $k=1$ (1); sum of zeroes $=-k=-1$ (1); other zero $=-1-(-2)=1$ (1). | 1+1+1 |
| 17 | Sum $=3$, product $=-10$ (1+1); polynomial $x^2-(3)x+(-10)=x^2-3x-10$ (1). | 1+1+1 |
| | **OR:** Zeroes $\frac{3}{2}$, $-2$ (1); sum $-\frac{1}{2}=-b/a$ (1); product $-3=c/a$ (1). | 1+1+1 |
| 18 | $p(4)=0$: $16-24+8=0$ gives $k=-6$ (1); sum of zeroes $=-k=6$ (1); other zero $=6-(4)=2$ (1). | 1+1+1 |
| | **OR:** Sum $=-11$, product $=24$ (1+1); polynomial $x^2-(-11)x+(24)=x^2+11x+24$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $\alpha+\beta=9$, $\alpha\beta=18$ (1+1); new sum $=11$ (1); new product $=\alpha\beta+(\alpha+\beta)+1=28$ (1); polynomial $x^2-(11)x+(28)$ (1). | 1+1+1+1+1 |
| | **OR:** Sum of zeroes $=-b/a=3$ (2); third zero $=3-(2)-(-2)=3$ (2); check by product $=-12$ (1). | 2+2+1 |
| 20 | $\alpha+\beta=5$, $\alpha\beta=-14$ (1+1); new sum $=7$ (1); new product $=\alpha\beta+(\alpha+\beta)+1=-8$ (1); polynomial $x^2-(7)x+(-8)$ (1). | 1+1+1+1+1 |
| | **OR:** Sum of zeroes $=-b/a=4$ (2); third zero $=4-(1)-(-3)=6$ (2); check by product $=-18$ (1). | 2+2+1 |

**Mistakes to avoid:**

- Writing sum of zeros as $\frac{c}{a}$ and product as $-\frac{b}{a}$; it is sum $=-\frac{b}{a}$ and product $=\frac{c}{a}$.
- Forgetting the sign: for $x^2-8x+15$ the sum of zeros is $+8$, not $-8$.
- Counting the zeros of $y=p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2+(\alpha+\beta)x+\alpha\beta$ instead of $x^2-(\alpha+\beta)x+\alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2+1$ has none).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 7 | Sum $= -b/a$ |
| 2 | **(b)** 4 | Each point where the graph meets the $x$-axis is a zero |
| 3 | **(a)** 7 | Sum of zeroes $= 11$, so other $= 11 - 4 = 7$ |
| 4 | **(b)** $x^2 - 3x - 18$ | $x^2 - (\text{sum})x + \text{product}$ |
| 5 | **(b)** -4 | $\alpha + \beta = -b/a$ |
| 6 | **(b)** -12 | $\alpha\beta = c/a$ |
| 7 | **(a)** 3 and 7 | $(x-3)(x-7) = 0$ gives $x = 3$ or $x = 7$ |
| 8 | **(c)** 8 | Sum $= -b/a$ |
| 9 | **(d)** 0 | Each point where the graph meets the $x$-axis is a zero |
| 10 | **(b)** 5 | Sum of zeroes $= 2$, so other $= 2 - -3 = 5$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $x^2 - (\frac{3}{2})x + (1)$ (1) $= x^2 - \frac{3}{2}x + 1$ (1). | 1+1 |
| 12 | Zeroes are 2 and 6 (1); a parabola, so degree 2 (1). | 1+1 |
| 13 | Sum $= -\dfrac{b}{a} = -2$ (1); product $= \dfrac{c}{a} = \frac{1}{5}$ (1). | 1+1 |
| 14 | $x^2 - (-4)x + (4)$ (1) $= x^2 + 4x + 4$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Zeroes $6, -1$ (1); sum $5 = -b/a$ (1); product $-6 = c/a$ (1). | 1+1+1 |
| 16 | $p(-3) = 0$: $9 - 6 - 3 = 0$ gives $k = 2$ (1); sum of zeroes $= -k = -2$ (1); other zero $= -2 - (-3) = 1$ (1). | 1+1+1 |
| 17 | Sum $= 11$, product $= 18$ (1+1); polynomial $x^2 - (11)x + (18) = x^2 - 11x + 18$ (1). | 1+1+1 |
| | **OR:** Zeroes $-\frac{1}{3}$, 3 (1); sum $\frac{8}{3} = -b/a$ (1); product $-1 = c/a$ (1). | 1+1+1 |
| 18 | $p(5) = 0$: $25 - 45 + 20 = 0$ gives $k = -9$ (1); sum of zeroes $= -k = 9$ (1); other zero $= 9 - (5) = 4$ (1). | 1+1+1 |
| | **OR:** Sum $= -5$, product $= -6$ (1+1); polynomial $x^2 - (-5)x + (-6) = x^2 + 5x - 6$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $\alpha + \beta = 10$, $\alpha\beta = 9$ (1+1); new sum $= 12$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = 20$ (1); polynomial $x^2 - (12)x + (20)$ (1). | 1+1+1+1+1 |
| | **OR:** Sum of zeroes $= -b/a = 4$ (2); third zero $= 4 - (-1) - (-2) = 7$ (2); check by product $= 14$ (1). | 2+2+1 |
| 20 | $\alpha + \beta = -1$, $\alpha\beta = -12$ (1+1); new sum $= 1$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = -12$ (1); polynomial $x^2 - (1)x + (-12)$ (1). | 1+1+1+1+1 |
| | **OR:** Sum of zeroes $= -b/a = 3$ (2); third zero $= 3 - (2) - (5) = -4$ (2); check by product $= -40$ (1). | 2+2+1 |

**Mistakes to avoid:**

- Writing sum of zeros as $\dfrac{c}{a}$ and product as $-\dfrac{b}{a}$; it is sum $= -\dfrac{b}{a}$ and product $= \dfrac{c}{a}$.
- Forgetting the sign: for $x^2 - 8x + 15$ the sum of zeros is $+8$, not $-8$.
- Counting the zeros of $y = p(x)$ as the y-intercept; zeros are where the graph meets the *x*-axis.
- Counting a parabola that only touches the x-axis as having no zeros or two different zeros; it has one repeated zero.
- Writing the polynomial from its zeros as $x^2 + (\alpha + \beta)x + \alpha\beta$ instead of $x^2 - (\alpha + \beta)x + \alpha\beta$.
- Claiming a quadratic polynomial always has two real zeros (e.g. $x^2 + 1$ has none).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

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